Development of a Task-Oriented Chatbot Application for Monitoring Taiwan Photon Source Front-End System

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- Major Project:
  - EPICS Application Development
  - TPS Front-End X-Ray Beam Position Monitor Data Analysis
  - Cloud Based IoT Platform Develpment
- Research Interest:
  - Internet of Things Application
  - Photonics System Design
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Motivation

A Long time ago
In a galaxy far, far away……

Motivation

~~A Long time ago~~

~~In a galaxy far, far away~~……

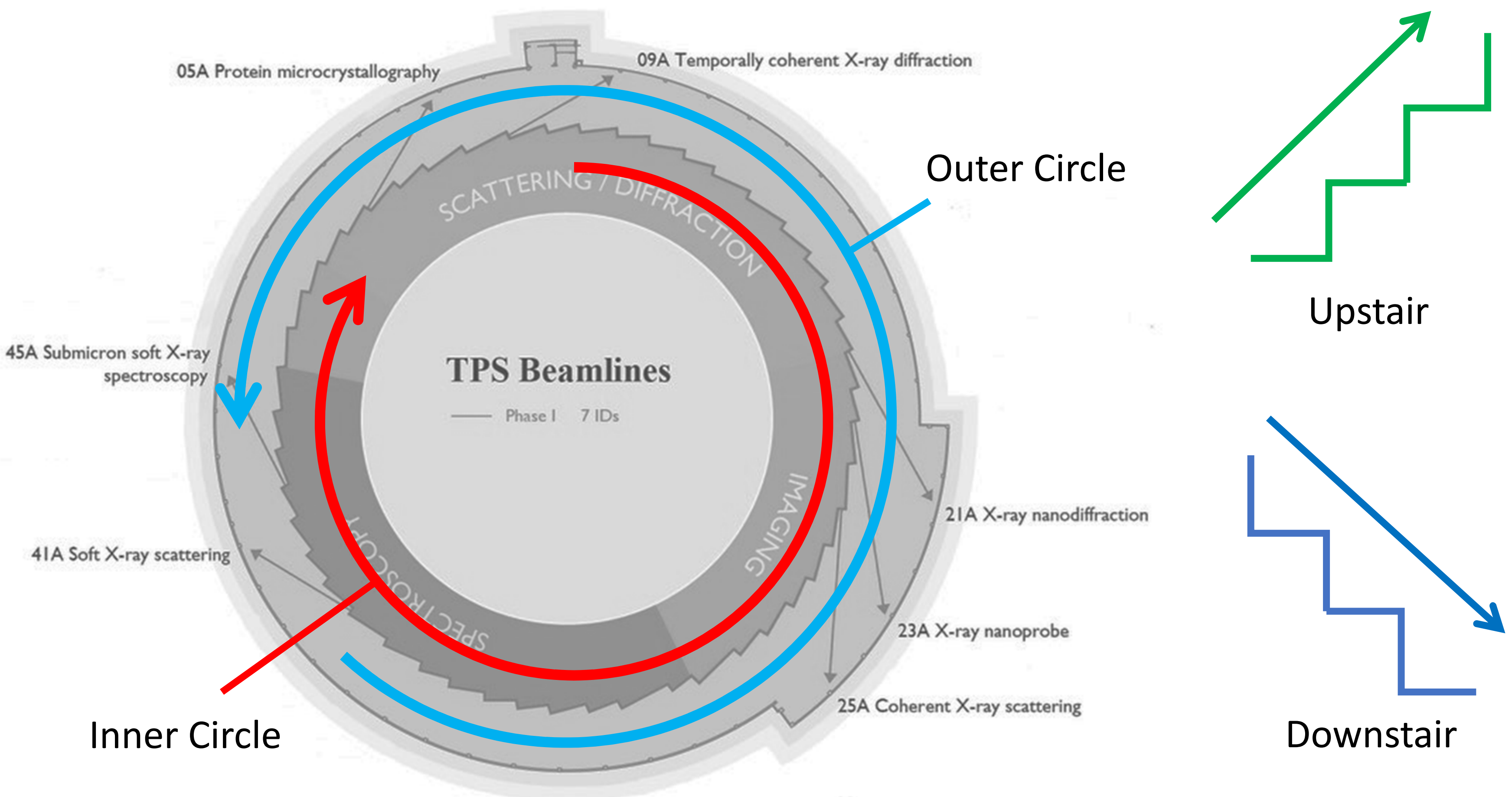
Motivation

~~A Long time ago~~
~~In a galaxy far, far away~~......

At the end of last year
We start a machine study project......

TPS Phase-1 All X-Ray Beam Position Monitors Scanning, Calibration and Analysis

This Project we need walk a lot……



After Several Machine Studies

I want to develop a new application
and I hope this application is easy to use.

Because……

- We need get machine status and scan status anywhere.
- Laptop is too heavy and not very convenience.

Conversational UI: Chatbot

Chatbot is a dialogue system.

SIRI

Google Assistant

LINE Message API

- LINE is most popular social network mobile APP in Taiwan
- LINE provide LINE Message API in 2016

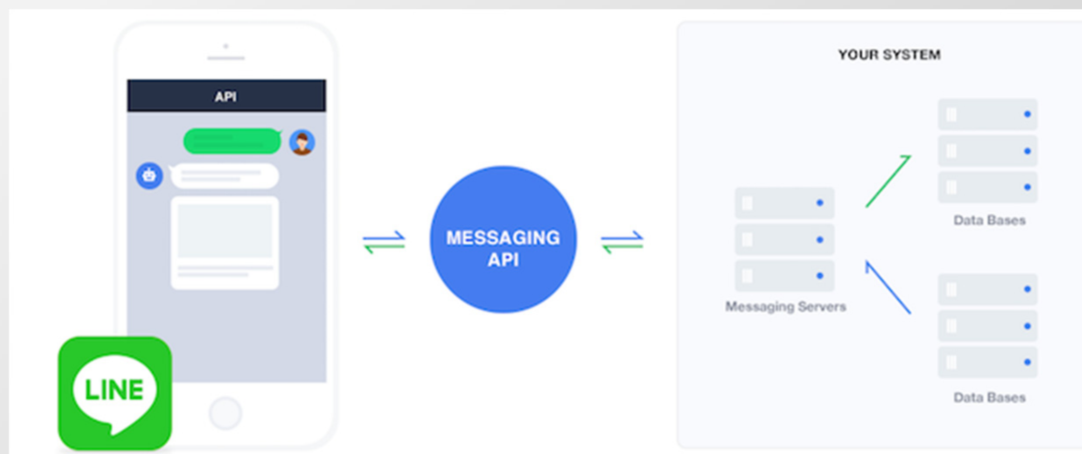
IOS App Store

Screenshot at 2018/10/18

LINE Message API

LINE official SDKs for the Messaging API:

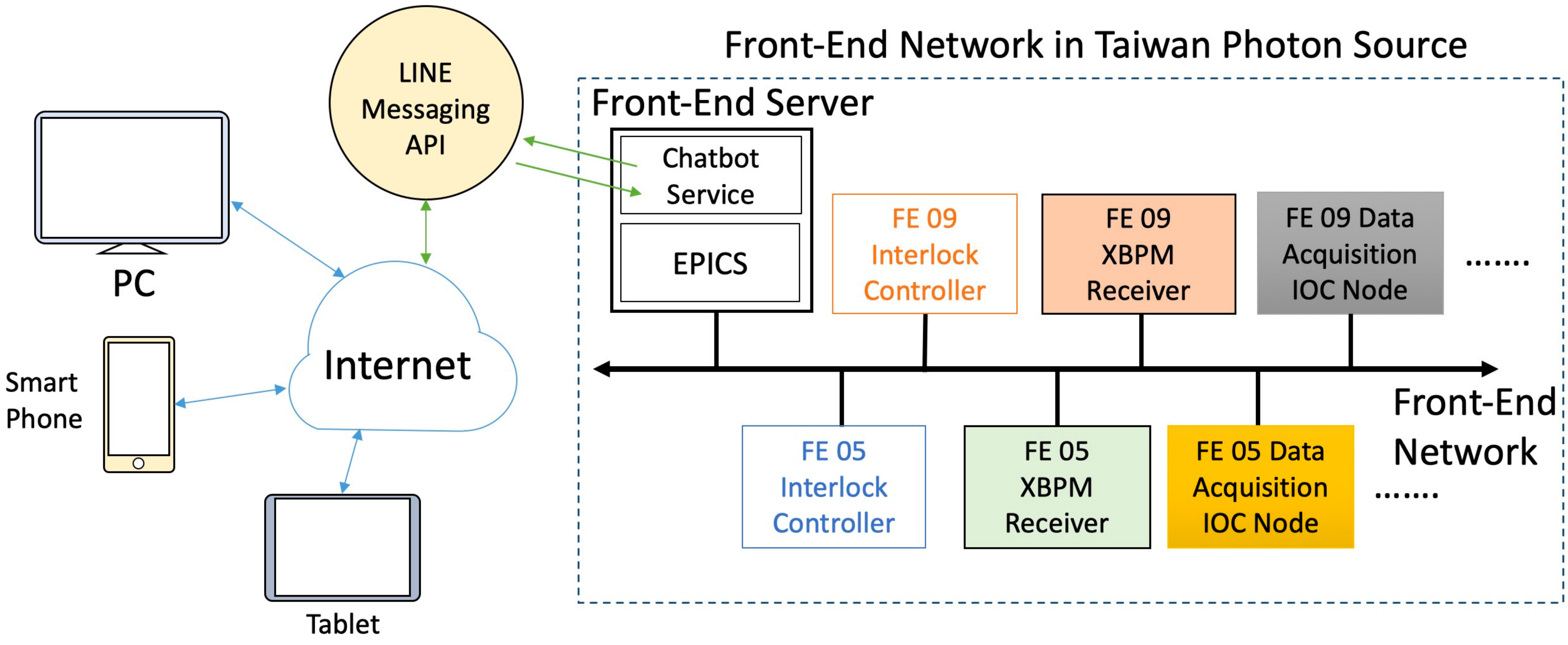
Java, PHP, Go, Perl, Ruby, Python, Node.js



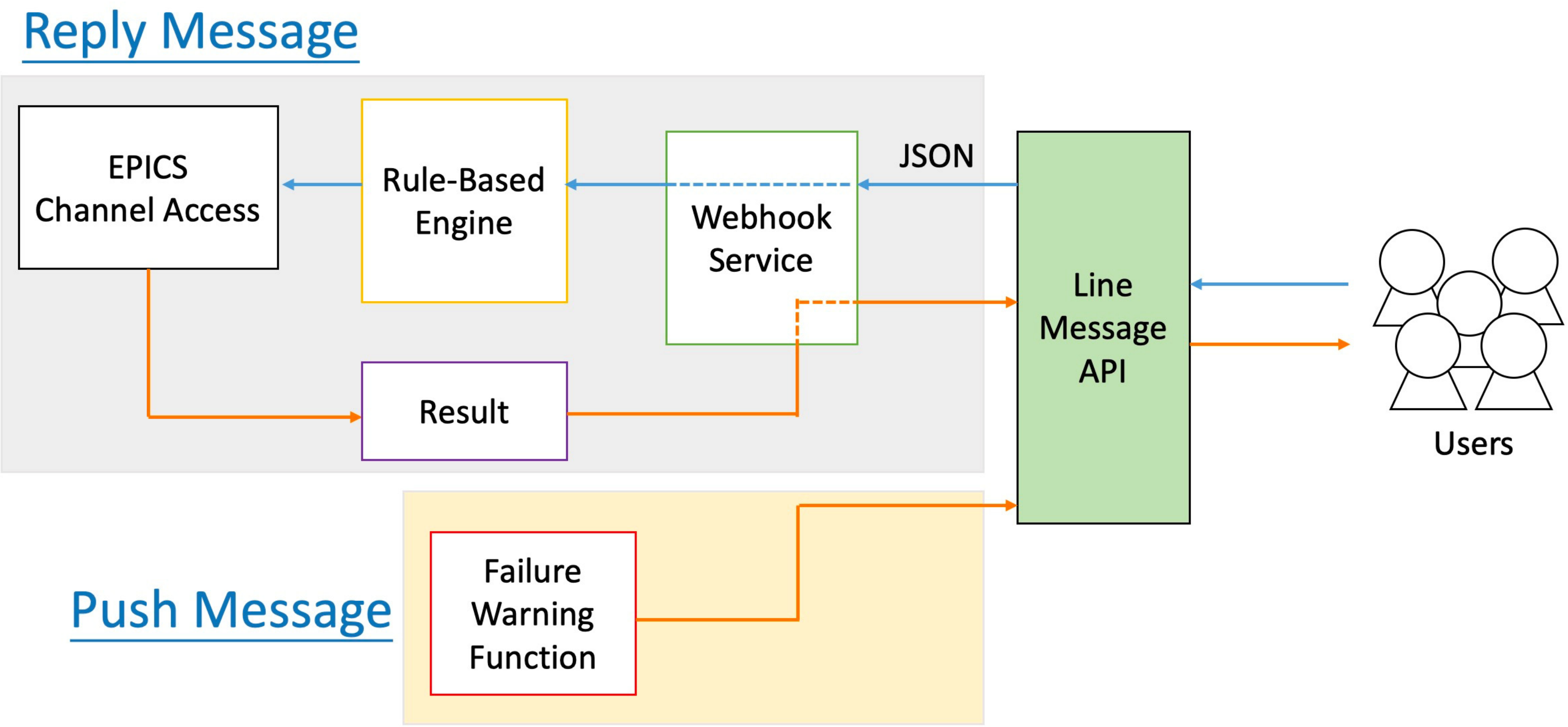
Credit: LINE Developer

https://developers.line.me/en/docs/messaging-api/overview/

NSRRC FE Chatbot Application Network Architecture



Software Architecture



Webhook Service

- Webhook is made by flask framework

  When user send a message to Chatbot Server will Receive a JSON File from Line Message API


```
{
    "events": [{
        "type": "message",
        "replyToken": "44[redacted]b",
        "source": {
            "userId": "U9[redacted]5a",
            "type": "user"
        },
        "timestamp": 1539659447846,
        "message": {
            "type": "text",
            "id": "8[redacted]2",
            "text": "XBPM FE21"
        }
    }]
}
```

Rule-Based Engine

Take rule of query XBPM status as an example.

"XBPM FE21"

"XBPM" AND "FE"

Is Keywords both in the string? —False→ Check another keyword

True

FE_Number="21"

Filter out FE number with regular expression. —False→ Do nothing

True

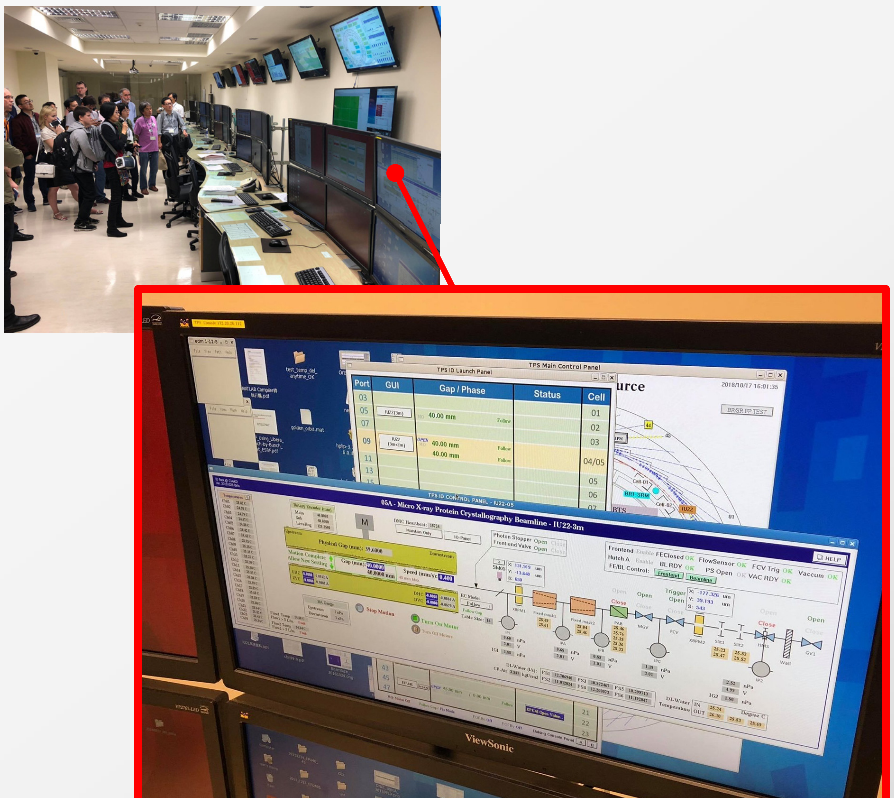
caget("FE-DI-XBPM-%s:signals:sa.X")%FE_Number

Use EPICS to get current value

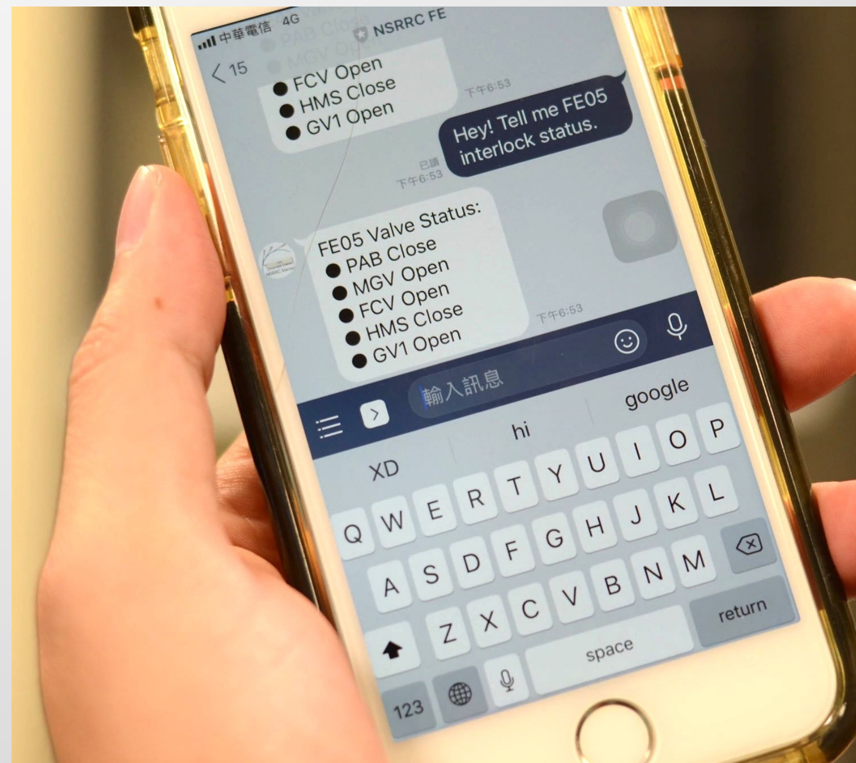
Graphic User Interface vs. Conversational User Interfaces

If I want to know FE05 valves current status…

- TPS Control Room GUI

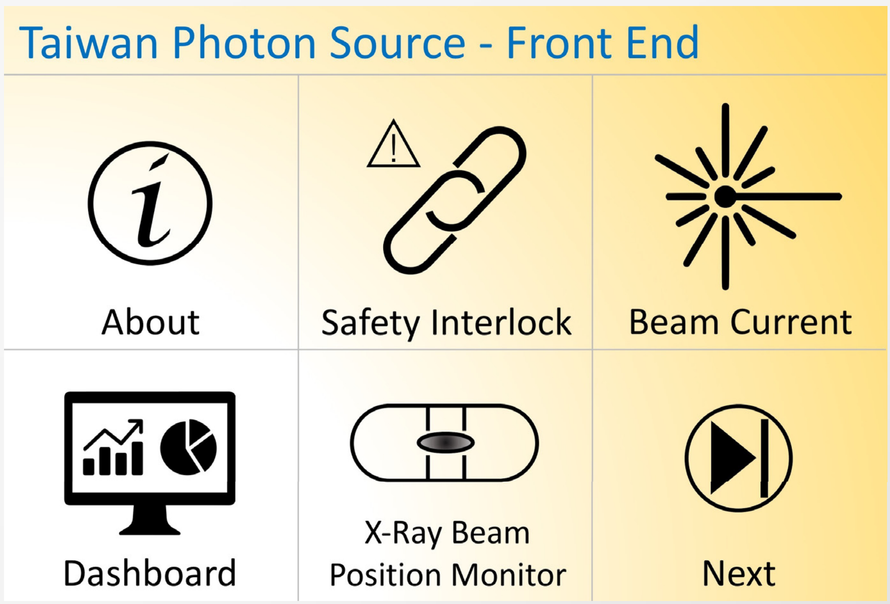


- My Chatbot Application

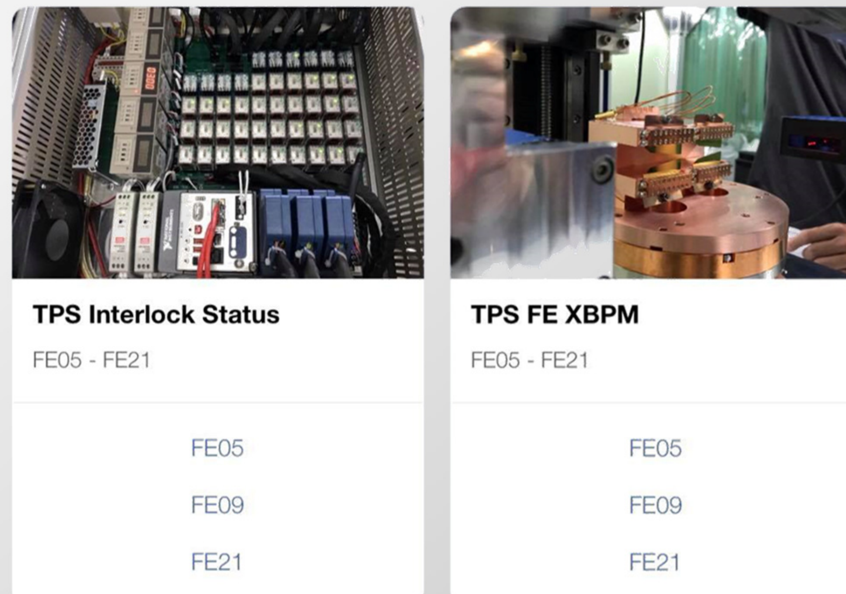


Make Chatbot Easier to Use

- Rich Menu



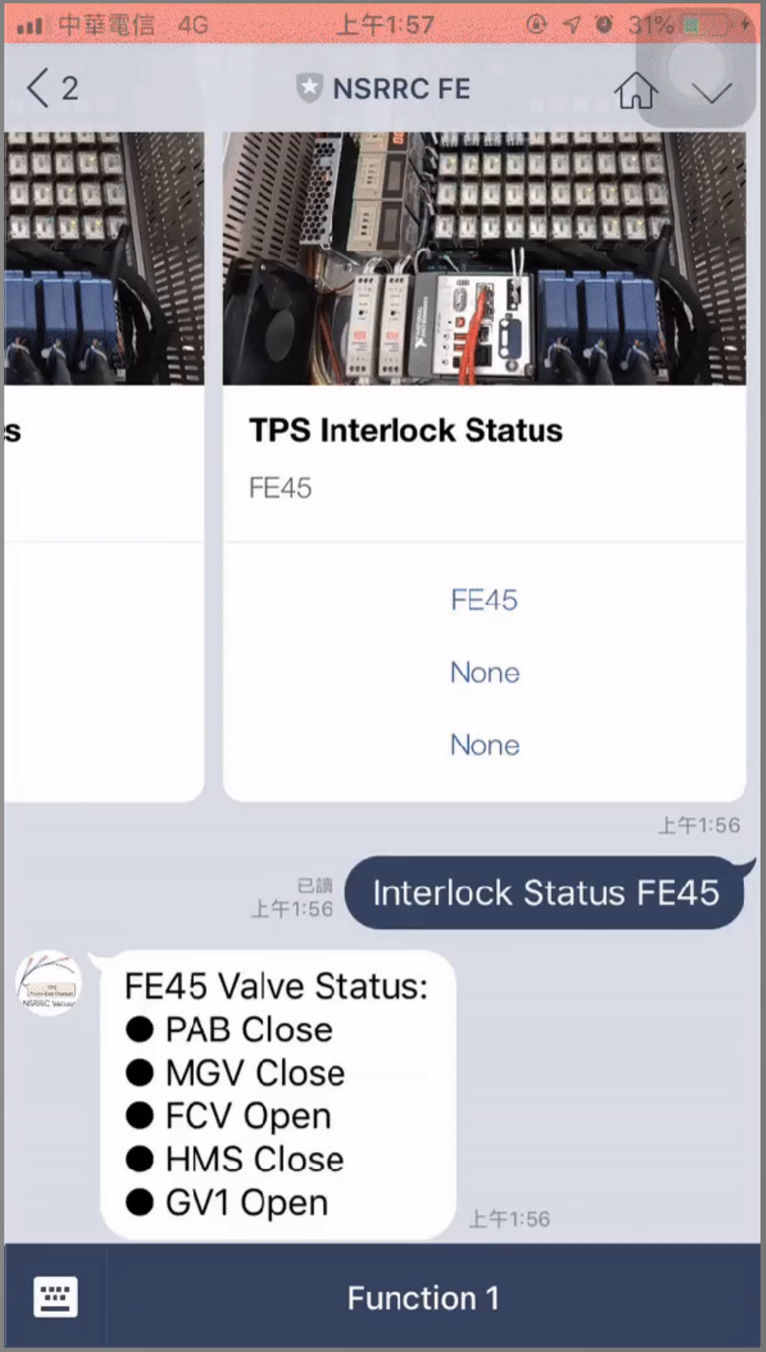
- Template type message



Demo

TPS Front-End Chatbot Application

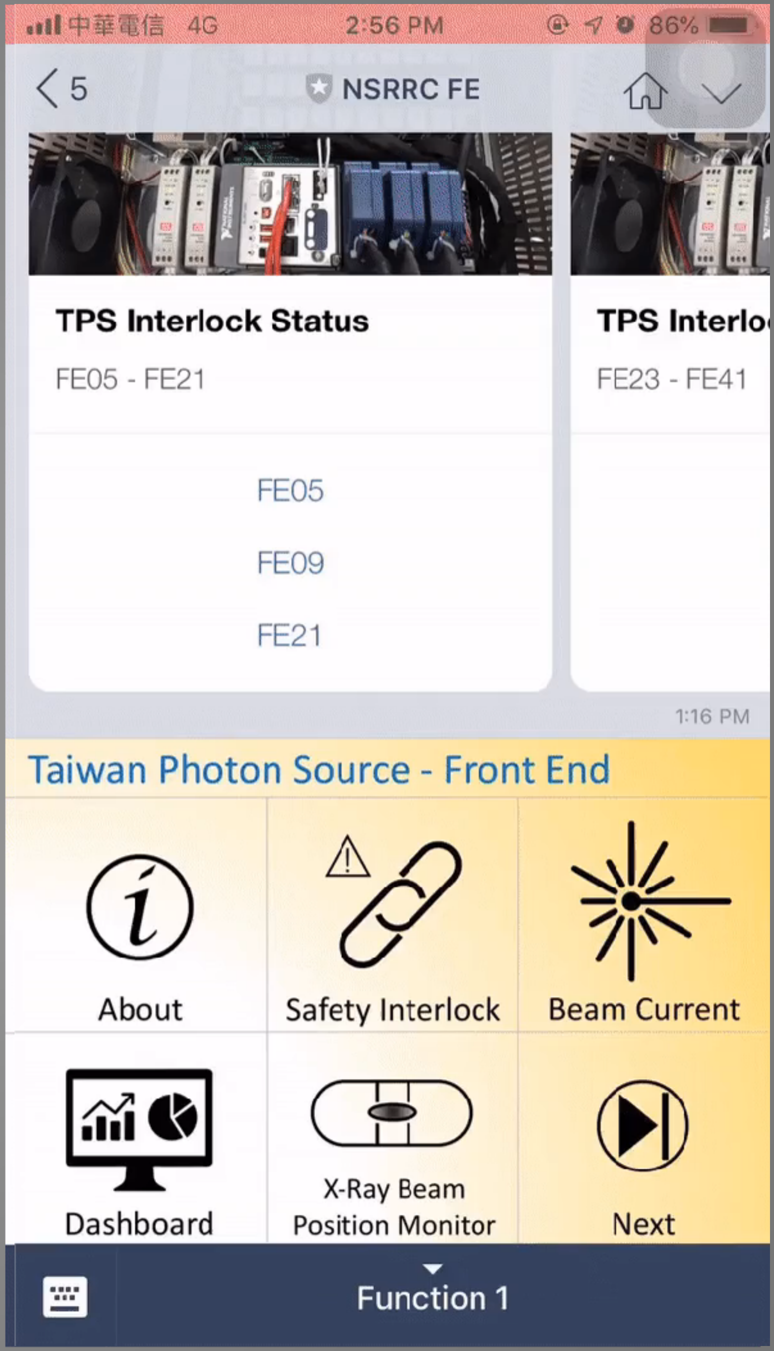
- Query Front-End Valve Status
- Query X-Ray Beam Position Monitor Value



Demo

TPS Front-End Real-Time Dashboard

Embed Grafana dashboard into Chatbot Application with LINE Front-end Framework



Summary

At Present

- The chatbot platform for TPS Front-End is finished.
- 127 EPICS PV that this chatbot can query.
- 133 EPICS PV are monitoring for failure warning.

Increasing the number of device that this chatbot can query and monitoring

In the future

- Make this chatbot support other platform: Facebook, WhatAPP, etc.
- Finish Grafana UI.

Thank You